QSCP Hamilton Inspection Department

Inspection Report

Site: Hamilton Plant

No.	Status	Date
QSCP-INS-2010-021	Issued	2010-05-10

External Visual Inspection - ATU-100 Piping

Scope and Responsibilities

This report details the 5-yearly external visual inspection of the piping systems within the ATU-100. The inspection was conducted to verify the external condition of the assets and ensure compliance with the site rules for Class 2 amine service. Douglas Ferreira served as the API 570 inspector responsible for the field assessment and technical review of the piping circuits.

The scope of work follows the requirements established in API 570 and API 574 for the periodic monitoring of pressure piping. Douglas Ferreira executed the visual survey in accordance with the QSCP-IP-ATU100 Rev 1, which governs the inspection intervals and CML requirements for this unit, including the specific 3-year interval for the regenerator overhead circuit.

Reference Documentation

The current inspection findings are compared against the baseline established in QSCP-INS-2005-019, which recorded the previous 5-yearly external visual inspection. Technical guidance for the assessment of insulation and support conditions was derived from API 574. All thickness evaluations and remaining life calculations are governed by the criteria defined in API 570.

The QSCP-IP-ATU100 Rev 1 remains the governing document for the circuitization of the unit. This plan incorporates the 138 total CMLs and the revised 3-year thickness interval for the regenerator overhead. The inspection also accounts for the previous replacement of a failed elbow as noted in the 2008-02-15 revision of the plan.

Inspection Findings - 4-RA-104

Line 4-RA-104, a rich-side L/R exchanger line constructed of A106 Gr.B, was inspected for external degradation. The line has a design pressure of 862 kPa(g) and a design temperature of 127 °C. During the survey, damaged jacketing and wet insulation were identified on several sections of this circuit. Given the potential for moisture entrapment against the carbon steel surface, a CUI inspection is recommended.

The nominal thickness of 8.56 mm and the retirement thickness of 3.81 mm serve as the basis for future thickness monitoring. Although this was a 5-yearly external visual, the presence of wet insulation necessitates targeted NDE to ensure the actual wall thickness has not degraded toward the t-min. The corrosion rate will be calculated per site rules using the larger of the long-term and short-term rates once new readings are obtained.

Inspection Findings - 4-LA-104

Line 4-LA-104, the lean-side L/R exchanger piping, was also assessed. This A106 Gr.B line is rated for a design pressure of 2758 kPa(g) and a design temperature of 138 °C. Similar to the rich-side piping, this line exhibited damaged jacketing and evidence of wet insulation. These conditions are known precursors for Corrosion Under Insulation in the ATU-100.

The t-min for 4-LA-104 is 3.18 mm against a nominal wall of 6.02 mm. Per the site rules, any reading below this retirement thickness must be raised as an NCR. Douglas Ferreira has recommended a follow-up CUI inspection to verify that the external corrosion has not compromised the pressure boundary. The next thickness reading for this Class 2 circuit must be scheduled at no more than half the remaining life.

Sign-off	Name	Function	Date	Signature
Prepared	Douglas Ferreira	Inspection Supervisor, API 570 inspector	2010-05-10	_____